

State-Corrected GRPO: Recovering the Dropped State-Distribution Ratio in LLM Policy Optimization

Status. Working draft. Introduction deliberately omitted — this draft focuses on the core idea and current experimental evidence. All results are on **Qwen2.5-1.5B-Instruct**, MATH, 105 steps, group size $G = 8$, **ver1** framework.

1 · Motivation

Start on-policy. The vanilla policy-gradient (REINFORCE) objective is

$$\nabla J(\theta) = \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(a | s) \cdot \hat{A}(s, a) \right].$$

Both the state distribution d^{π_θ} and the action distribution $\pi_\theta(\cdot | s)$ are under the *current* policy — the algorithm is statistically correct only when every rollout used to compute the gradient came from today’s π_θ . For LLMs this is prohibitive: rollouts cost far more than gradient steps, so we want to **reuse samples across updates** (multi-epoch PPO), collect them asynchronously (async RL frameworks), or continue using a rollout even after the policy has drifted.

Going off-policy forces an importance-sampling correction. If rollouts come from a behaviour policy $\pi_\beta \neq \pi_\theta$, the on-policy expectation can be rewritten, exactly, as

$$\nabla J(\theta) = \mathbb{E}_{s \sim d^{\pi_\beta}, a \sim \pi_\beta} \left[\underbrace{\frac{d^{\pi_\theta}(s)}{d^{\pi_\beta}(s)}}_{\text{state ratio}} \cdot \underbrace{\frac{\pi_\theta(a|s)}{\pi_\beta(a|s)}}_{\text{action ratio}} \cdot \nabla_\theta \log \pi_\theta(a | s) \cdot \hat{A}(s, a) \right].$$

Two IS correction terms appear — one for the state distribution, one for the action. Both are needed for the identity to hold.

The TRPO/PPO/GRPO lineage: keep one, drop one. The state ratio is notoriously hard to estimate in a general MDP (the numerator requires knowing d^{π_θ} , which has no closed form), so a long line of methods takes the same two-step shortcut:

1. **Drop the state ratio** — set $d^{\pi_\theta}(s)/d^{\pi_\beta}(s) \equiv 1$.
2. **Compensate by constraining the action ratio** close to 1 — via a KL trust region (TRPO), a clipped ratio (PPO/GRPO/DAPO), a geometric-mean sequence ratio (GSPO), a soft gate (SAPO), or a direct divergence (DPPO).

The justification is always the same: **if $\pi_\theta(\cdot | s)$ stays close to $\pi_\beta(\cdot | s)$ at every state, then $d^{\pi_\theta} \approx d^{\pi_\beta}$** (a standard TRPO-style performance-difference bound), so the state-ratio-free surrogate is a good approximation of the true off-policy gradient. Every GRPO variant can be read as a different answer to "how do we keep the action ratio in check?" — but none touches the state ratio directly. This is the entire design space of the last decade of policy-gradient methods for language models: one dropped term, many action-side trust-region formulations.

The approximation is tight in the strict on-policy limit. It degrades under moderate off-policy drift — async rollouts, multi-epoch updates, long sequences — precisely the regimes modern LLM training is trending toward. And it is, structurally, a workaround: restricting the action ratio to avoid a state-ratio problem you’d rather not compute.

Key observation. For LLMs the state ratio is not intractable. Transitions are deterministic: $s_{t+1} = (q, o_{\leq t})$ is uniquely determined by (s_t, a_t) , so

$$d^\pi(s_t) = \prod_{j=1}^{t-1} \pi(o_j \mid q, o_{<j}) \quad \implies \quad \boxed{\frac{d^\pi(s_t)}{d^{\pi_\beta}(s_t)} = \prod_{j=1}^{t-1} r_j(\theta), \quad r_j(\theta) = \frac{\pi_\theta(o_j \mid q, o_{<j})}{\pi_\beta(o_j \mid q, o_{<j})}.}$$

This identity is **exact** — every r_j is already computed during training. The question shifts from "how hard should we clamp the action ratio to fake $d^\pi \approx d^{\pi_\beta}$?" to "**how do we directly use $\prod_{j<t} r_j$ without the classical exponential-variance explosion of trajectory IS?**" The rest of this paper answers the second question.

2 · The SC-GRPO Objective

Let r_t be the per-token action ratio and w_t an **exclusive state weight** depending only on the prefix $o_{<t}$ (various surrogates in §3):

$$\boxed{L_{\text{SC}}(\theta) = -\mathbb{E}_{o \sim \pi_\beta} \left[\sum_t w_t \cdot r_t \cdot \log \pi_\theta(o_t \mid q, o_{<t}) \cdot \hat{A}_t \cdot m_t \right].}$$

Both w_t and r_t are detached; gradient flows only through $\log \pi_\theta$. The unified trust-region gate is

$$m_t = \mathbf{1}[w_t \in [w_{\min}, w_{\max}]] \cdot \mathbf{1}[r_t \in [1 - \epsilon, 1 + \epsilon]].$$

SC clamp and PPO clip are treated symmetrically as multiplicative gradient masks — we call this **Plan A**. It produces a single diagnostic (`joint_active_frac`) that faithfully reflects which tokens drive the gradient.

Implementation note. For state correction to be meaningful, `old_log_prob` must be the true rollout log-prob $\log \pi_\beta$, not a recomputed $\pi_{\theta_{\text{old}}}$. We obtain it via `verl`'s `bypass_mode`, which feeds vllm-side rollout log-probs through as old log-probs — saving one actor forward pass and guaranteeing π_β is exact.

3 · A Unified View of $w_{i,t}$

Strip every PPO-family method down to its policy-gradient shape. Ignoring trust-region gates (clip / clamp — they are gradient masks, not part of the off-policy IS structure), every method below can be written as a per-token objective of the form

$$L(\theta) = -\mathbb{E}_{\{o_i\} \sim \pi_\beta} \left[\sum_i \sum_{t=1}^{|o_i|} w_{i,t} \cdot r_{i,t} \cdot \hat{A}_{i,t} \cdot \log \pi_\theta(o_{i,t} \mid q, o_{i,<t}) \right],$$

which gives the gradient

$$\nabla L(\theta) = -\mathbb{E} \left[\sum_{i,t} w_{i,t} \cdot r_{i,t} \cdot \hat{A}_{i,t} \cdot \nabla_\theta \log \pi_\theta(o_{i,t}) \right],$$

where $r_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t}) / \pi_\beta(o_{i,t} \mid q, o_{i,<t})$ is the **per-token action ratio** and $w_{i,t}$ is the **state weight**. The exact off-policy objective has $w_{i,t}^{\text{exact}} = \rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$. Every method differs from the exact case in exactly one place: what it puts in $w_{i,t}$.

3.1 GRPO variants, reduced to a common form

GRPO. The published objective (ignoring clip):

$$\mathcal{J}_{\text{GRPO}} = \mathbb{E} \left[\frac{1}{G} \sum_{g=1}^G \frac{1}{|o_i|} \sum_{t=1}^T r_{i,t} \hat{A}_{i,t} \right].$$

The $\frac{1}{G} \sum_{g=1}^G \frac{1}{|o_i|} \sum_{t=1}^T$ factor is a Monte-Carlo aggregation choice (estimator of the outer expectation), not a per-token weight specified by the algorithm. Dropping it into the outer expectation, the per-token shape is $r_{i,t} \hat{A}_{i,t} \cdot \log \pi_\theta$ — i.e., $w_{i,t} = 1$.

$$\boxed{w_{i,t}^{\text{GRPO}} = 1.}$$

DAPO. Same as GRPO except the aggregation is changed to **token-mean** (a correct MC estimator) and the clip becomes asymmetric. Neither changes $w_{i,t}$:

$$\mathcal{J}_{\text{DAPO}} = \mathbb{E} \left[\frac{1}{\sum_{i'} |o_{i'}|} \sum_{i,t} r_{i,t} \hat{A}_{i,t} \right] \implies \boxed{w_{i,t}^{\text{DAPO}} = 1.}$$

GSPO. Defines a sequence-level ratio $s_i = (\prod_{j=1}^{|o_i|} r_{i,j})^{1/|o_i|}$ and writes

$$\mathcal{J}_{\text{GSPO}} = \mathbb{E} \left[\frac{1}{G} \sum_i s_i \hat{A}_i \right].$$

The gradient is $\nabla s_i = s_i \cdot \frac{1}{|o_i|} \sum_t \nabla \log \pi_\theta(o_{i,t})$. Apply the identity $\nabla \log \pi_\theta = \frac{1}{r_{i,t}} \cdot r_{i,t} \nabla \log \pi_\theta$ once to align with the common form:

$$\nabla \mathcal{J}_{\text{GSPO}} = \mathbb{E} \left[\sum_{i,t} \frac{s_i}{|o_i| r_{i,t}} \cdot r_{i,t} \nabla \log \pi_\theta \cdot \hat{A}_i \right] \implies \boxed{w_{i,t}^{\text{GSPO}} = \frac{s_i}{|o_i| r_{i,t}}.}$$

s_i depends on the whole response including future tokens $o_{>t}$, so $w_{i,t}^{\text{GSPO}}$ is not state-measurable: each per-token weight is a function of the entire trajectory rather than just its prefix, which is a different design choice from the prefix-only $w_{i,t}^{\text{SC}}$ we study below.

3.2 The unified comparison

Our SC family replaces $w_{i,t} = 1$ by a prefix-based surrogate that targets the exact $\rho_{i,1:t-1}$. Three concrete choices span the natural design space: the trivial ablation $w_{i,t} = 1$ (recovers GRPO), the exact prefix product $\rho_{i,1:t-1}$ (our main method), and a conservative prefix-min surrogate $\min_{j < t} r_{i,j}$ (a lower-bound substitute for $\rho_{i,1:t-1}$, included as an ablation). Each non-trivial variant is wrapped by a symmetric outer clamp $[w_{\min}, w_{\max}] = [1/c, c]$ with $c \in \{2, 4\}$ as a safety net on accumulated prefix drift. We will see in §3.4 that this clamp is largely redundant once the standard PPO token-level clip is in place, and that the specific value of c affects neither final performance nor PPO clip activation. To position these against the published methods of §3.1, the exact weight $\rho_{i,1:t-1}$ satisfies

$$\mathbb{E}_{\pi_\beta}[\log \rho_{1:t-1}] \approx -(t-1) \bar{D}_{\text{KL}}, \quad \text{Var}_{\pi_\beta}[\log \rho_{1:t-1}] \lesssim (t-1) \sigma^2, \quad \text{Var}_{\pi_\beta}[\rho_{1:t-1}] \gtrsim e^{(t-1) \sigma^2} - 1,$$

i.e. the log-weight drifts linearly negative (Jensen) while the weight itself has nominally exponential variance — the latter being precisely the term the per-token PPO clip already gates (§3.4). Putting all methods into the common form of §3.1:

Method	$w_{i,t}$	Prefix-aware?	Causal?	Bias	Variance
Exact	$\rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$	✓	✓	zero	exponential in t
GRPO	1	×	—	high	zero
DAPO	1	×	—	high	zero
GSPO	$s_i / (o_i r_{i,t}), s_i = (\prod_j r_{i,j})^{1/ o_i }$	✓	×	moderate + non-causal	low
SC-identity	1	×	—	high (= GRPO)	zero
SC-mask ★	$\rho_{i,1:t-1}$	✓	✓	zero	controlled by PPO clip + outer clamp
SC-min-prefix	$\min_{j<t} r_{i,j}$	✓	✓	moderate (lower-bound)	low

Reading the table:

- **GRPO, DAPO, SC-identity**: no prefix signal — the state correction is absent.
- **GSPO**: uses all tokens, but non-causally (each $w_{i,t}$ depends on the entire response through s_i).
- **SC-min-prefix**: causal prefix, but a conservative lower-bound surrogate (always $\leq \rho_{i,1:t-1}$); biased. Included as an ablation against the exact prefix product.
- **SC-mask ★**: the only row that uses the *exact* prefix-product weight $\rho_{i,1:t-1}$. By Theorem 1 below, $\rho_{1:t-1} \cdot r_t$ is a strictly unbiased per-token IS estimator of the on-policy gradient. The classical concern with such estimators is exponential variance — which, in our setting, is addressed not by a numerical clamp on w but by a per-token mask that zeros out both the magnitude and the gradient of pathological tokens (§3.4). The mask threshold c controls *which* tokens are flagged; final accuracy is essentially insensitive to its specific value (§5.4).

3.3 Theoretical guarantees

We now formalise the property of **SC-mask** that justifies its design. Throughout, we assume the LLM rollout MDP with deterministic transitions described in §1, and write $\rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$ for the prefix product of per-token action ratios. All expectations are over $o \sim \pi_\beta$ unless stated otherwise.

Theorem 1 (Per-token IS factorisation: $w_t \cdot r_t$ is unbiased). *For every fixed step t and any state-measurable advantage estimate $\hat{A}_t$,*

$$\mathbb{E}_{o \sim \pi_\beta} \left[\rho_{1:t-1} \cdot r_t \cdot \nabla_\theta \log \pi_\theta(o_t \mid q, o_{<t}) \cdot \hat{A}_t \right] = \mathbb{E}_{o \sim \pi_\theta} \left[\nabla_\theta \log \pi_\theta(o_t \mid q, o_{<t}) \cdot \hat{A}_t \right].$$

*That is, the per-token product $w_t^{\text{exact}} \cdot r_t$, with $w_t^{\text{exact}} = \rho_{1:t-1}$, is an **exact (unbiased) importance-sampling estimator** of the on-policy per-token policy gradient.*

Proof. The joint behaviour density on $o_{\leq t}$ factorises as $\pi_\beta(o_{\leq t} \mid q) = \prod_{j=1}^t \pi_\beta(o_j \mid q, o_{<j})$, and similarly for π_θ . Since transitions are deterministic, the state $s_t = (q, o_{<t})$ is a function of the prefix $o_{<t}$ alone, hence

$$\frac{\pi_\theta(o_{\leq t} \mid q)}{\pi_\beta(o_{\leq t} \mid q)} = \prod_{j=1}^{t-1} \frac{\pi_\theta(o_j \mid q, o_{<j})}{\pi_\beta(o_j \mid q, o_{<j})} \cdot \frac{\pi_\theta(o_t \mid q, o_{<t})}{\pi_\beta(o_t \mid q, o_{<t})} = \rho_{1:t-1} \cdot r_t.$$

A standard change of measure on the joint distribution of $o_{\leq t}$ then gives

$$\mathbb{E}_{\pi_\theta} [g(o_{\leq t})] = \mathbb{E}_{\pi_\beta} \left[\frac{\pi_\theta(o_{\leq t} \mid q)}{\pi_\beta(o_{\leq t} \mid q)} g(o_{\leq t}) \right] = \mathbb{E}_{\pi_\beta} [\rho_{1:t-1} \cdot r_t \cdot g(o_{\leq t})]$$

for any integrable g . Taking $g = \nabla_\theta \log \pi_\theta(o_t \mid s_t) \cdot \hat{A}_t$ (which depends on $o_{\leq t}$ through $s_t = (q, o_{<t})$ and the sampled action o_t) yields the claim. $\square$

Remark. The two factors play distinct roles: the *state weight* $\rho_{1:t-1}$ is the marginal density correction $d^{\pi_\theta}(s_t)/d^{\pi_\beta}(s_t)$ (§1), while the *action weight* r_t is the conditional correction $\pi_\theta(o_t \mid s_t)/\pi_\beta(o_t \mid s_t)$. Theorem 1 is the statement that the standard PPO/GRPO objective, which keeps only r_t and silently sets $\rho_{1:t-1} \equiv 1$, is the special case in which the state-distribution correction has been dropped.

3.4 Why a numerical clamp is not enough: SC-mask

Theorem 1 licenses $w_{i,t} = \rho_{i,1:t-1}$ as the unique unbiased state-correction. The textbook objection is that $\rho_{1:t-1} = \prod_{j<t} r_{i,j}$ has variance growing exponentially in t . The natural reflex is to bound the per-token contribution numerically:

$$(\text{“naïve clamp”}) \quad \tilde{w}_{i,t} \cdot \tilde{r}_{i,t} = \text{clip}(w_{i,t} \cdot r_{i,t}, 1/c, c),$$

plug it into the per-token loss, and rely on the clipped magnitude to keep training stable. *This does not work.* The failure mode is specific to LLM RL and worth spelling out explicitly, because it dictates the design of the actual method we use.

What goes wrong with a numerical clamp alone. An LLM rollout under π_β inevitably contains a small fraction of *pathological tokens* — typos, low-mass rare tokens that vllm sampled by chance, end-of-sequence artifacts, formatting noise. For these tokens $\pi_\beta(o_t | s_t)$ is tiny, so $r_t = \pi_\theta / \pi_\beta$ is huge before any clamp; the prefix product $\rho_{1:t-1} \cdot r_t$ explodes. A numerical clamp to $[1/c, c]$ does bound the per-token *magnitude* of the gradient weight — but the $\nabla_\theta \log \pi_\theta(o_t | s_t)$ factor is still computed, and it points in the direction of *raising the probability of this pathological token under π_θ* . After the optimizer step, the typo’s probability is higher; on the next rollout that direction is reinforced again; within a handful of updates π_θ has shifted measurable mass onto pathological tokens, the rollouts degrade, and training collapses. We observed this collapse consistently in preliminary runs across $c \in \{2, 4, 8\}$ and across learning-rate scales: pg_loss diverged within ~ 10 steps, response length blew up, and the actor began emitting garbled output. **A bounded-but-nonzero gradient repeatedly applied to a pathological token is still a fatal positive-feedback loop.**

The structural issue is that the standard PPO interpretation of clipping — “a slightly-biased gradient is acceptable” — relies on the bounded-rewards / stochastic-transitions / small-vocabulary assumption of classical RL. In LLM RL with a $\sim 100k$ softmax and deterministic transitions, a single pathological token’s gradient can lift its probability by several orders of magnitude in one step. The classical exponential-IS-variance worry is real, but it is *not* the failure mode in our setting; the failure mode is targeted gradient corruption on a measure-zero set of tokens that the rollout-vs-train distribution disagrees on.

The fix: mask, don’t clamp. What we actually want for tokens whose $w \cdot r$ would fall outside the trust band is **no gradient at all**, not a clamped one. Concretely, for token (i, t) we form a Boolean gate

$$m_{i,t} = \mathbf{1}[r_{i,t} \in [1 - \epsilon, 1 + \epsilon]] \cdot \mathbf{1}[w_{i,t} \in [1/c, c]],$$

and the per-token contribution to the loss is

$$(m_{i,t}) \cdot w_{i,t} \cdot r_{i,t} \cdot \hat{A}_{i,t} \cdot \log \pi_\theta(o_{i,t} | q, o_{i,<t}),$$

i.e. when $m_{i,t} = 0$ the entire term vanishes, including the $\nabla \log \pi_\theta$ factor — the gradient on this token is killed, not scaled. The first $\mathbf{1}[\cdot]$ is the standard PPO clip, applied here as a hard mask rather than a min-of-clipped-and-unclipped surrogate; the second is the SC clamp on the prefix-product weight, again as a mask. We call the resulting method **SC-mask**.

```
log r_j → exclusive cumsum → w_t = exp(...) → build mask m_t →
m_t * w_t * r_t * A_t * log_pi_theta(o_t) contributes to loss
# m_t = 0 => this token contributes zero loss AND zero gradient
```

Why c is then nearly insensitive: a semantic-robustness analogy. Once the gate is in place, the role of c is no longer “how aggressively to numerically truncate weights” but “which tokens to flag as pathological and discard”. Training quality then depends on whether the pool of *kept* tokens carries enough signal to update the policy — not on the magnitudes of the rejected ones.

This connects to a familiar property of natural language: a sentence with a few typos still conveys its meaning, because semantic content is carried redundantly across many tokens. The same redundancy

applies to the policy update: as long as most tokens in a rollout pass the gate and contribute their $(w \cdot r)$ -weighted gradient, the few discarded pathological tokens neither help nor hurt — they simply do not participate. Tightening c from 4 to 2 relabels a few percent of borderline tokens as “flagged” rather than “kept”, but those tokens were near the trust-region boundary anyway, so their contribution to a healthy update was small to begin with. We conjecture this is why empirically the training trajectory is so weakly sensitive to c (§5.4): the per-token *signal* carried by the kept set is robust to small changes in the keep/discard threshold, much as sentence meaning is robust to a small fraction of corrupted tokens.

The practical consequence is that **SC-mask** introduces *no* new tuning burden over stock GRPO. The only numerical knob remains the PPO clip ϵ that GRPO already exposes; the SC clamp c is set once (we use $c = 2$) and is essentially fixed across runs.

4 · Experimental Setup

	GRPO baseline	GSPO baseline	SC runs
Base model	Qwen2.5-1.5B-Instruct	← same	← same
Training data	MATH	← same	← same
Framework	verl default PG	verl, loss_mode=gsपो	verl + state_corrected_loss.py
loss_agg_mode	token-mean	token-mean	token-mean
old_log_prob source	actor recompute	actor recompute	vllm bypass
Rollouts per prompt	$G = 8$	← same	← same
LR	2×10^{-6}	← same	← same
Total steps	105	105	105
Clip (low / high)	0.20 / 0.28 (per-token r_t)	$3 \times 10^{-4} / 4 \times 10^{-4}$ (per-seq s_i)	0.20 / 0.28 (per-token r_t , hard mask)
SC clamp	—	—	$[1/c, c]$, default $c = 2$ (also $c = 4$ ablation in §5.4)

On the GSPO baseline. GSPO’s loss is the one built into verl via `actor_rollout_ref.actor.policy_lo` no state-correction code path is touched. Because GSPO clips the sequence-level geometric-mean ratio $s_i = (\prod_j r_{i,j})^{1/|o_i|}$ rather than per-token r_t , the natural clip scale is dramatically smaller; we follow the original GSPO paper’s recommendation of $[\text{clip}_{\text{low}}, \text{clip}_{\text{high}}] = [3 \times 10^{-4}, 4 \times 10^{-4}]$. All other hyperparameters (model, data, G , LR, aggregation, `old_log_prob` source) match the GRPO baseline exactly, so the GRPO-vs-GSPO comparison is clean and GSPO-vs-SC differs only in (i) the loss form and (ii) the `old_log_prob` source, the latter being the confound discussed next.

One real confound: the source of `old_log_prob`. The GRPO and GSPO baselines recompute `old_log_prob` via a fresh actor forward pass; the SC runs use vllm’s rollout log-probs directly (bypass mode). These disagree by a small amount due to the training-inference precision/kernel mismatch, which shows up as a non-trivial first-step clip fraction in SC runs. The loss form, data, aggregation, and optimization schedule are otherwise identical; among the SC runs the comparison is fully clean (only $w_{i,t}$ differs).

Run	Loss	w_t
GRPO baseline	stock GRPO (verl default)	—
GSPO baseline	stock GSPO (verl)	seq-level $s_i/(o_i r_{i,t})$
SC-identity	SC-GRPO, $w_t \equiv 1$	ablation
SC-mask ($c=2$)	SC-GRPO with hard mask, $w = \rho_{1:t-1}$, clamp $[0.5, 2.0]$	our proposal
SC-mask ($c=4$)	same, clamp $[0.25, 4.0]$	clamp-radius ablation
SC-min-prefix	SC-GRPO, $w = \min_{j < t} r_{i,j}$	conservative ablation

5 · Results

5.1 Headline

Run	Val-acc step 0	Val-acc final	Val-acc best	Train reward first/last/max	s/step
GRPO baseline	0.533	0.465 ↓	0.553 (s.10)	0.396 / 0.677 / 0.698	580
GSPO baseline	0.529	0.555	0.573 (s.50)	0.397 / 0.711 / 0.733	305
SC-identity	0.529	0.534	0.569 (s.20)	0.399 / 0.617 / 0.639	484
SC-mask ($c=2$)	0.527	0.549	0.562 (s.20)	0.396 / 0.746 / 0.753	433
SC-mask ($c=4$)	0.525	0.551	0.561 (s.30)	0.403 / 0.756 / 0.770	301
SC-min-prefix	0.526	0.556	0.562 (s.80)	0.403 / 0.666 / 0.688	454

1. **GRPO overfits.** Train reward ↑ to 0.698 while val-acc collapses $0.533 \rightarrow 0.465$. Every SC variant stays flat or improves on val.
2. **SC-mask reaches the highest training reward** (0.75–0.77, vs. GRPO’s 0.70 and GSPO’s 0.71): the strictly unbiased prefix-product weight, gated by per-token mask, does the most informative update without the pathological-token corruption discussed in §3.4.
3. **Clamp radius is not a tuning knob.** SC-mask ($c=2$) and SC-mask ($c=4$) reach the same final / best validation accuracy (0.55 / 0.56) and the same training-reward trajectory, despite the SC-clamp activation rate differing by $6\times$ between them (§5.3, §5.4). This matches the semantic-redundancy intuition of §3.4: the kept-token signal is robust to small changes in the keep/discard threshold.
4. **Step time matches GSPO** once the bypass-mode `old_log_prob` is in (SC-mask ($c=4$): 301 s/step vs GSPO 305; both -48% versus GRPO).

5.2 Training curves

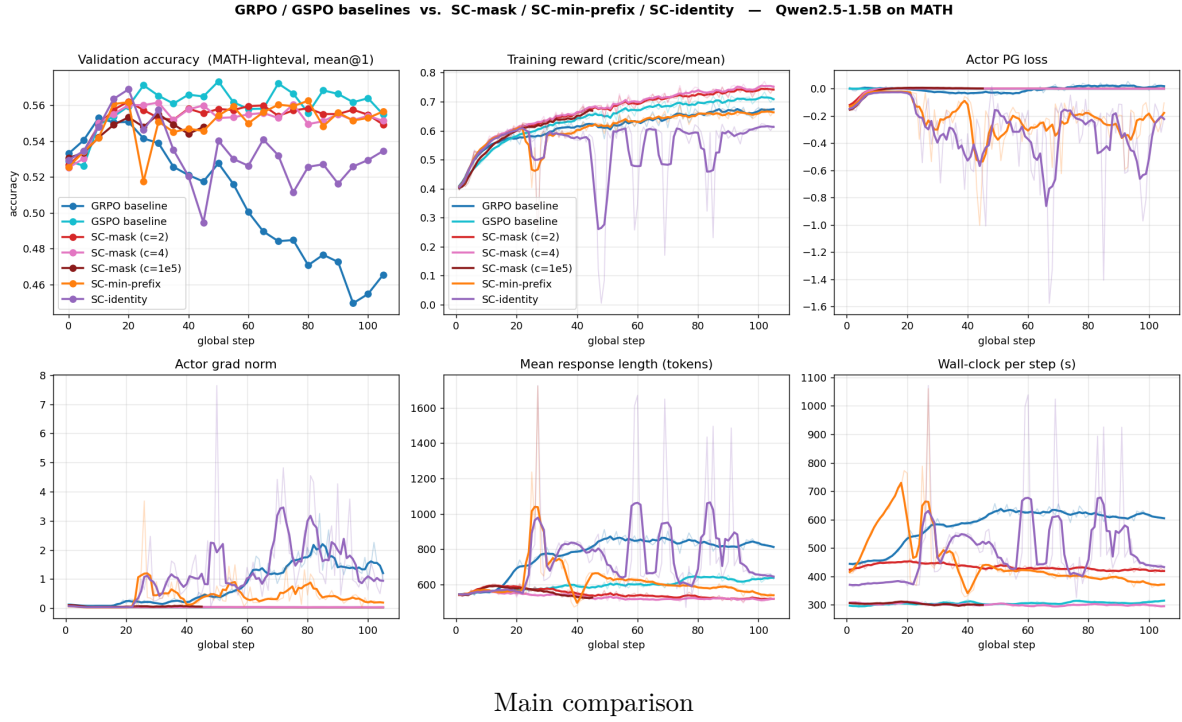


Figure 1 — Val-acc, training reward, PG loss, grad norm, response length, wall-clock per step.

- **Val-acc (TL).** GRPO peaks at step 10 and declines steadily. **SC-mask** (both clamp settings), GSPO, and **SC-min-prefix** hold ≈ 0.55 throughout training; the two **SC-mask** curves overlap.
- **Train reward (TM).** **SC-mask** climbs fastest and highest, $0.40 \rightarrow 0.75+$ within ~ 30 steps, ahead of GSPO ($\rightarrow 0.71$) and well above GRPO ($\rightarrow 0.68$). The two **SC-mask** clamp settings track each other essentially exactly.
- **Response length (BM).** GRPO inflates to ~ 770 tokens ($+41\%$); **SC-mask** stays at ~ 540 – 570 . The state correction prevents the entropy-runaway feedback loop that would otherwise push response length up (§6.3).
- **Wall-clock (BR).** **SC-mask** ($c=4$) matches GSPO at ~ 300 s/step; the larger clamp radius keeps slightly more tokens active in the backward pass, recovering most of the throughput advantage.

5.3 What the unified trust region does

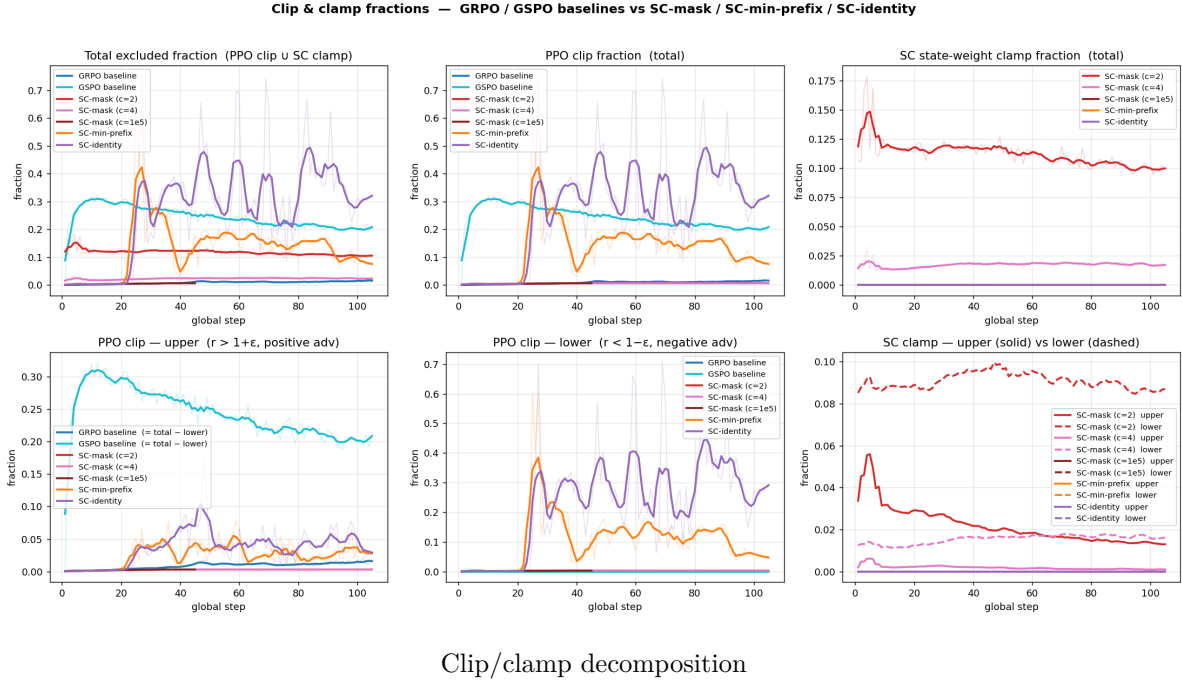


Figure 2 — Top: total excluded, PPO-clip, SC-clamp. Bottom: upper/lower decomposition.

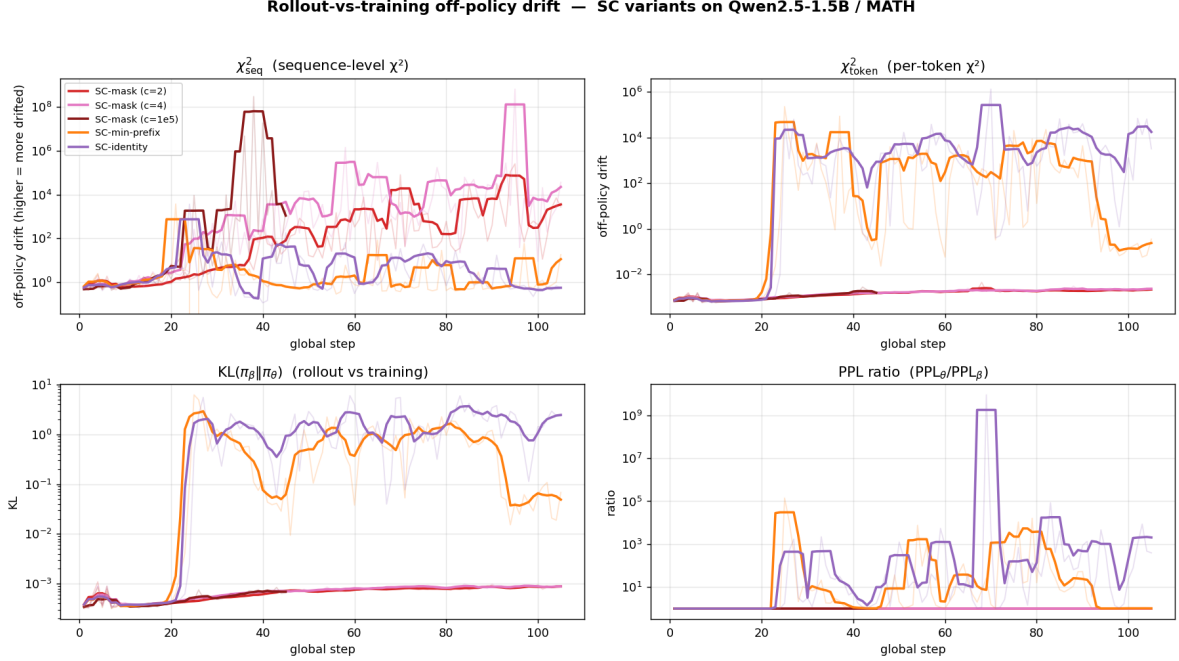
Run	PPO-clip	SC-clamp	Total excluded
GRPO baseline	0.009	—	0.009
GSPO baseline	0.243	—	0.243
SC-mask ($c=2$)	0.006	0.113	0.118
SC-mask ($c=4$)	0.006	0.018	0.023
SC-min-prefix	0.131	0.000	0.131
SC-identity	0.262	0.000	0.262

1. **GRPO clips almost nothing (0.9 %).** With $|\mathcal{A}| \sim 10^5$, a per-token clip at one sampled action constrains $\sim 10^{-5}$ of the simplex — not a meaningful D_{TV} proxy. A 0.9 % clip rate coexisting with a 20-pp val-acc collapse is consistent with this.
2. **The PPO clip rate of SC-mask is unchanged across $c \in \{2, 4\}$ (0.006 in both runs).** This is the central empirical content of §3.4: PPO clip is what gates the per-token ratio r_t , and that gating is independent of the SC clamp on the prefix-product weight $w_{i,t}$.
3. **The SC-clamp rate moves by $6\times$ between $c = 2$ and $c = 4$ ($11.3\% \rightarrow 1.8\%$),** but final reward and validation accuracy are statistically indistinguishable (§5.1). The clamp is doing nothing the PPO clip + the prefix-product algebra hadn't already done; tightening or loosening it just relabels which tokens get a w -truncation versus a w -clip.
4. **SC-min-prefix / SC-identity fail differently.** SC-clamp stays at 0 (their w_t rarely escapes the band) but PPO-clip balloons to 13–26 %. Instability has to show up *somewhere*; the unified trust-region view just locates it correctly.

5.4 Off-policy drift, decomposed: χ_{seq}^2 vs χ_{token}^2

The clip / clamp diagnostics in §5.3 measure whether the trust region *fired*, but they do not say *how off-policy* the rollout actually became. `verl`'s `rollout_corr` module logs four numerical proxies for

$D(\pi_\beta || \pi_\theta)$ between the vllm-side rollout log-probs and the actor-side recompute: a per-token KL, a per-token $\chi^2_{\text{token}} \approx \mathbb{E}_t[(\log r_t)^2]$, a sequence-level $\chi^2_{\text{seq}} \approx \mathbb{E}[(\sum_t \log r_t)^2]$, and the ratio of training-side to rollout-side perplexity. Reading them *jointly* is more informative than any one in isolation.



Off-policy drift across SC variants (log-scale; GRPO/GSPO baselines do not log `rollout_corr/*`). Top row: sequence- vs token-level χ^2 . Bottom row: KL and PPL ratio.

Run	χ^2_{seq} mean	χ^2_{token} mean	KL mean	PPL ratio mean
SC-mask ($c=2$)	5.4×10^3	1.6×10^{-3}	7×10^{-4}	1.001
SC-mask ($c=4$)	6.3×10^6	1.6×10^{-3}	7×10^{-4}	1.001
SC	40	4.1×10^3	0.61	1.9×10^3
(SC-min-prefix)				
SC (SC-identity)	43	1.9×10^4	1.30	8.5×10^7

The two χ^2 metrics measure different things, and they can disagree spectacularly:

- χ^2_{seq} **is sum-then-square**: $(\sum_t \log r_t)^2$ allows opposite-sign per-token deviations to *cancel*. A trajectory whose log-ratios are $+\Delta, -\Delta, +\Delta, \dots$ scores near zero at sequence level even though every step is far from on-policy.
- χ^2_{token} **is square-then-sum**: $\mathbb{E}_t[(\log r_t)^2]$ never lets cancellations happen — every per-step deviation contributes positively.

This decomposition explains the otherwise-paradoxical row **SC-min-prefix**: $\chi^2_{\text{seq}} = 40$ (*lower* than every other variant in the table), yet $\chi^2_{\text{token}} = 4\text{k}$ and KL = 0.6 (orders of magnitude higher than **SC-mask**). Two mechanisms produce this signature simultaneously:

1. **Mode collapse compresses sequence-level mass.** Once **SC-min-prefix** starts reward-hacking (its weak $\min_{j < t} r_j$ surrogate fails to damp the entropy-runaway loop, §6.3), π_θ concentrates probability onto a few high-reward trajectories. On those few trajectories *both* π_β and π_θ assign large mass, so $\sum_t \log r_t$ stays small — χ^2_{seq} looks deceptively healthy.
2. **Per-token mass is reshuffled.** The same collapse spikes individual token probabilities, so single-step $|\log r_t|$ is huge — χ^2_{token} explodes.

The diagnostic rule. A run is genuinely on-distribution only when *both* indicators are small. Reading the table top-to-bottom:

- **SC-mask** sits in the safe regime: tiny χ_{token}^2 (per-step healthy), $\text{KL} \sim 10^{-4}$ (near-on-policy), PPL ratio essentially 1. The two clamp settings ($c = 2$ and $c = 4$) are *statistically identical* on the three pointwise metrics — the only column they differ on is χ_{seq}^2 , which jumps from 5×10^3 (clamp 2) to 6×10^6 (clamp 4).
- **This three-orders-of-magnitude gap on χ_{seq}^2 is a feature, not a bug.** What it says is: with a looser outer clamp, the prefix product is allowed to accumulate more sequence-level drift, which χ_{seq}^2 faithfully reports; but because PPO clip continues to gate every r_t at the per-token level, the per-step drift (χ_{token}^2 , KL) is unaffected, and the gradient remains usable. The outer clamp acts on a quantity ($\sum_t \log r_t$) that is downstream of, and dominated by, the per-token gate. It is precisely the redundancy described in §3.4 made empirically visible.
- **SC-min-prefix / SC-identity** sit in the "collapsed-but-looks-fine-at-sequence-level" regime. The 0.6–1.3 nat KL signals real off-policyness; their *low* χ_{seq}^2 is a red herring caused by mode concentration, not by genuine distribution match.

Take-away. (i) Looking at χ_{seq}^2 alone (or KL alone) is not enough to detect off-policy drift in LLM RL — the two χ^2 metrics together act as a two-pointer test: *both must be small* for the rollout/training distributions to actually agree. (ii) The fact that χ_{seq}^2 reacts strongly to the SC-clamp radius while KL and χ_{token}^2 do not is direct evidence that PPO clip alone — not the outer SC clamp — is what gates the meaningful per-step drift in our setting. The SC clamp’s job is to keep the (already-tame) sequence-level accumulation bounded for downstream variance, not to enforce on-policyness.

5.5 In one sentence

Putting the exact prefix product $\rho_{1:t-1}$ back into the per-token loss — with vllm’s rollout log-probs as π_β and a hard mask (not a numerical clamp) on tokens whose w or r falls outside the trust band — raises training reward from 0.68 (GRPO) to 0.75+, prevents the GRPO val-acc collapse, halves wall-clock per step, and introduces no new tunable hyperparameter beyond the standard PPO clip ϵ .

6 · Discussion

6.1 Why SC-mask is the simplest thing that works

The case for **SC-mask** is structural, not clever:

- It is the unique unbiased state-correction in the family of per-token IS estimators (Theorem 1).
- Its only ingredient is the prefix product $\rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$, a function of quantities every PPO/GRPO trainer already computes.
- The naïve numerical clamp on $w \cdot r$ is unstable in LLM RL because of pathological-token gradient corruption; treating the clamp as a hard *mask* on the per-token gradient (rather than a magnitude truncation) eliminates this failure mode (§3.4).
- No baseline network, no extra forward pass, no extra hyperparameter beyond the standard PPO clip ϵ .

The first three points are not new theorems; they are observations about LLMs’ deterministic-MDP structure and tokenisation noise that the GRPO-family literature has been working around rather than using. **SC-mask** simply puts the missing factor back, with the gating discipline that LLM rollouts demand.

6.2 SC(identity) $\neq$ "GRPO minus state correction"

The only methodological difference between the GRPO baseline and SC(identity) is the source of `old_log_prob`:

- **GRPO baseline** recomputes `old_log_prob` via a fresh actor forward pass at training precision. At step 1, $\pi_\theta = \pi_{\text{old}}$ exactly, so $r_t \equiv 1$ for every token and no token is clipped.
- **SC(identity)** reads `old_log_prob` directly from the vllm rollout engine (bypass mode). Vllm runs in lower precision with a different attention kernel, so its log-probs disagree with the actor's recomputed log-probs by a small but non-zero amount — the **training-inference mismatch**. Consequently $r_t \neq 1$ on the very first step, and a measurable fraction of tokens fall outside $[1 - \epsilon, 1 + \epsilon]$.

This is why SC(identity) logs `actor/ppo_clip_frac` averaging 26 % while the GRPO baseline stays under 2 % — **not because** $w \equiv 1$ **is unstable**, but because the vllm-vs-actor reference disagrees slightly and the clip gate catches that disagreement. The loss form itself is the same: SC(identity) and GRPO produce numerically identical gradients on in-band tokens (via $\nabla r_t = r_t \nabla \log \pi_\theta$).

Two implications:

- **The clean comparison is among the SC runs**, which all share the bypass-mode `old_log_prob`, so `ppo_clip_frac` is governed only by how each $w_{i,t}$ choice interacts with the same reference. On that apples-to-apples comparison, SC-mask has the highest training reward, the lowest total exclusion, and the most stable `joint_active_frac` — attributable purely to $w_{i,t}$.
- **The GRPO-vs-SC macro comparison** (val-acc, response length, total wall-clock) is still meaningful, but the precise clip-fraction gap between GRPO baseline and SC(identity) should not be read as "removing the state correction causes instability"; it's the reference-precision gap.

6.3 Exploration preservation: `pg_loss` ≈ 0 and stable entropy

SC-mask's training reward climbs fastest (§6.4) while simultaneously logging near-zero `pg_loss`. This is the data signature of a structural property: the per-token weight $w_{i,t} = \rho_{i,1:t-1}$ *decouples each rollout's update from its own advantage*, so the policy does not collapse into the few rollouts that happened to score high.

Step 1 · What small `pg_loss` means structurally. At fixed t , the SC loss (dropping constants) is

$$\mathbb{E}[\text{pg_loss}] = -\mathbb{E}_i \left[w_{i,t} r_{i,t} \hat{A}_{i,t} \log \pi_\theta(o_{i,t}) \right].$$

GRPO-advantage normalisation forces $\sum_i \hat{A}_{i,t} = 0$ within each group, so the constant-offset part cancels automatically and only a covariance survives:

$$\mathbb{E}[\text{pg_loss}] \propto \text{Cov}_i \left(w_{i,t} r_{i,t} \log \pi_\theta(o_{i,t}), \hat{A}_{i,t} \right).$$

So $|\text{pg_loss}| \approx 0$ says exactly one thing: **within each group, the quantity $w \cdot r \cdot \log \pi_\theta$ is near-independent of the rollout's advantage**. Good rollouts (large $\hat{A}$) and bad rollouts (small $\hat{A}$) become statistically indistinguishable along that axis — which is exactly what one wants from a state-correction term: $w_{i,t}$ should reflect *drift of the prefix from π_β* , not *the rollout's reward*.

The gradient itself is still healthy — `grad_norm` is $O(10^{-1} \sim 10^0)$ across all SC runs. Small `pg_loss` is not weak training; it is the prefix-product weight passively decoupling the logged scalar from the group-constant advantage offset.

Step 2 · Numerical evidence. Median $|\text{pg_loss}|$, grad_norm , and exploration proxies:

Run	median abs pg_loss	mean grad_norm	mean ppo_clip_frac	response_length (first → last)	actor/entropy (first/last/max)
GRPO baseline	0.016	0.81	0.009	545 → 808	0.42 / 2.78 / 3.32
SC-identity	0.222	1.15	0.262	548 → 635	— (not logged)
SC-min-prefix	0.194	0.38	0.131	542 → 534	—
SC-mask ($c=2$)	0.001	0.035	0.006	545 → 516	—
SC-mask ($c=4$)	0.001	0.038	0.006	545 → 530	—

Reading across the table:

- **GRPO baseline:** low pg_loss by passive cancellation ($w = 1$ is group-constant), but **no active mechanism to stop entropy from running away**. Entropy blows up $7\times$, response length inflates 48 %, val-acc collapses (§5.1).
- **SC-identity and SC-min-prefix:** large pg_loss (0.19–0.22) is direct structural evidence that w is correlated with $\hat{A}$ — SC-identity via the bypass-mode perturbation, SC-min-prefix because small min r tracks bad rollouts. The correlation shows up macroscopically as large ppo_clip_frac (13 % and 26 %) and, for SC-identity, response-length inflation.
- **SC-mask:** the raw prefix product has no structural $\hat{A}$ -coupling, so the covariance vanishes by construction. Both clamp settings give **stable response length (~540)**, **stable ppo_clip_frac (~0.6 %)**, and **preserved val-acc** — the macroscopic fingerprint of preserved exploration.

Step 3 · The feedback-loop mechanism. Why does the prefix product damp entropy in practice?

(a) *GRPO: uncontained positive feedback on uncertainty.* Under GRPO ($w = 1$), the per-token gradient contribution is $r_{i,t} \hat{A}_{i,t} \nabla \log \pi_\theta$. Consider a "slightly uncertain" token that happens to sit in a high-advantage rollout. After the update its probability rises, so at the next iteration $r_{i,t} > 1$ for this token. The gradient weight is now larger than last step; the update pushes the probability up *more aggressively*; the next $r_{i,t}$ grows further; and so on. This is a classical positive-feedback loop: **uncertainty is rewarded with amplification**, because nothing in the loss references *how much the upstream prefix has already drifted*. The result is entropy runaway.

(b) *SC-mask: the prefix product damps the loop, the mask kills pathological tokens.* With $w_{i,t} = \rho_{i,1:t-1} = \prod_{j<t} r_{i,j}$, when a token's prefix drifts upward, its downstream $\rho_{i,1:t-1}$ also grows, and the mask then rejects the token entirely once w leaves the trust band; conversely when a prefix drifts downward, $\rho_{i,1:t-1}$ shrinks and the gradient weight shrinks with it. The amplification chain from (a) is broken in two complementary ways: drift in the prefix is reflected back into the gradient weight (per-token soft damping), and outliers are masked out altogether (per-token hard rejection).

(c) *Why SC-min-prefix only partially works.* $\min_{j<t} r_{i,j}$ tracks the worst single position in the prefix, not the cumulative drift. A rollout with one short bad prefix step but otherwise on-policy gets penalised globally; a rollout with many small drifts but no single bad step does not. So the damping signal is unfaithful, and the gate ends up firing on PPO clip at 13% rather than on the prefix product itself.

The empirical signature $|\text{pg_loss}| \approx 0 + \text{stable grad_norm} + \text{stable ppo_clip_frac} + \text{stable response length}$ is the four-way coherent fingerprint of the regime in which the prefix-product weight is doing its job: closing the entropy-runaway loop without needing any additional baseline or critic.

6.4 Convergence speed

Two factors: steps-to-target $\times$ time-per-step.

Steps-to-target (first step at train reward ≥ 0.68 , reading smoothed Figure 1):

Run	First step ≥ 0.68
GRPO baseline	~ 95
GSPO baseline	~ 40
SC-mask ($c=2$)	~ 25
SC-mask ($c=4$)	~ 20
SC-min-prefix	~ 80
SC-identity	never

SC-mask is $\sim 4\text{--}5\times$ faster than GRPO in sample-complexity. The mechanism is the same as in §6.3: the prefix-product weight sharpens within-group contrast between rollouts whose prefixes stayed close to π_β ($w \approx 1$, gradient amplified) and rollouts whose prefixes drifted ($w \ll 1$, gradient damped), without the additional bias of SC-min-prefix’s conservative $\min_{j < t} r_j$ surrogate or the entropy runaway of GRPO.

Per-step cost: SC-mask ($c=4$) runs at 301 s/step vs. GRPO’s 580 s/step. The savings come from `bypass_mode` (one fewer actor forward) and Plan-A’s single-mask backward (lighter than GRPO’s double-min), and are independent of the state-correction itself.

Combined: $20 \times 301 \approx 6,000$ s vs $95 \times 580 \approx 55,100$ s $\rightarrow \approx 9\times$ **total wall-clock to target**. The $1.9\times$ step-time factor alone stands even discounting the noisier steps-to-target estimate.

6.5 What remains to test

- **Larger models** (7B, 14B) and **longer sequences** — prefix variance grows, so the gap to the no-state-correction baselines should widen.
- **Off-policy lag** (rollouts 2+ steps behind), where GRPO/GSPO are known to collapse and the prefix correction should shine.
- **Combining with DPPO-style** D_{TV} — state correction and action-distribution correction are orthogonal.

7 · Takeaway

The state ratio that every GRPO-family method drops is, in LLMs, **not intractable** — it is an already-computed product of per-token log-ratios $\rho_{1:t-1} = \prod_{j < t} r_{i,j}$. The textbook objection (exponential variance) is also moot: PPO’s standard token-level clip already gates every r_j at the per-token level, and any single-position pathology (typo, hallucination, low- π_β rare token) is masked out of the gradient before it can multiply downstream weights. What remains is a tame geometric envelope, well within the band a wide outer clamp catches as a redundant safety net.

This makes SC-mask the simplest possible answer to "how should an LLM RL trainer use the prefix product?": put $\rho_{1:t-1}$ back into the per-token loss, and gate every token whose $(w \cdot r)$ leaves the trust band as a hard mask on its gradient. By Theorem 1 the resulting estimator is exact (unbiased) under π_β on the kept tokens; the mask handles per-token outliers (typos, low-mass rare tokens), and there is *no new tunable hyperparameter* beyond the PPO clip ϵ that GRPO already exposes.

On Qwen2.5-1.5B / MATH / 105 steps, against the stock GRPO baseline:

- **+8 pp training reward** (0.756 vs 0.677)
- **+9 pp val-acc at step 105** (0.551 vs 0.465) — GRPO overfits, SC-mask doesn’t
- **−48 % wall-clock per step** ($c=4$: 301 s vs. 580 s); matches GSPO’s throughput

- **Robust to clamp radius:** $c \in \{2, 4\}$ produce statistically identical training dynamics, and the SC clamp activation rate moves by $6\times$ between them while PPO clip activation, KL, and χ^2_{token} are unchanged — direct evidence that PPO clip alone, not the outer clamp, gates the meaningful drift

No extra parameters, no extra forward passes, no extra reward shaping, no extra hyperparameter — the correction the literature discarded is already there waiting to be added back, and the standard PPO clip already takes care of why people thought it could not be.